

HAT-P-27b

R-0080

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-27_250416 · created 2026-07-15T22:21:23+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460781.8691 +/- 0.0085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.04
Transit depth (Rp/Rs)^2	1.91 +/- 1.1 [%]
Orbital Inclination (inc)	88.73 +/- 2.84
Airmass coefficient 1 (a1)	0.041 +/- 0.023
Airmass coefficient 2 (a2)	2.73 +/- 1.09
Scatter in the residuals of the lightcurve fit is	54.21 %
Best Comparison Star	#3 - [469, 441]
Optimal Aperture	2.88
Optimal Annulus	4.75
Transit Duration (day)	0.082 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.04 vs 0.119 (deviation 0.48σ)
Duration fit vs published	0.082 d vs 0.07 d (deviation 0.8σ)
Mid-time O–C	-20.0 min
Depth SNR	3.5
Residual scatter	54.21 %

Light curve

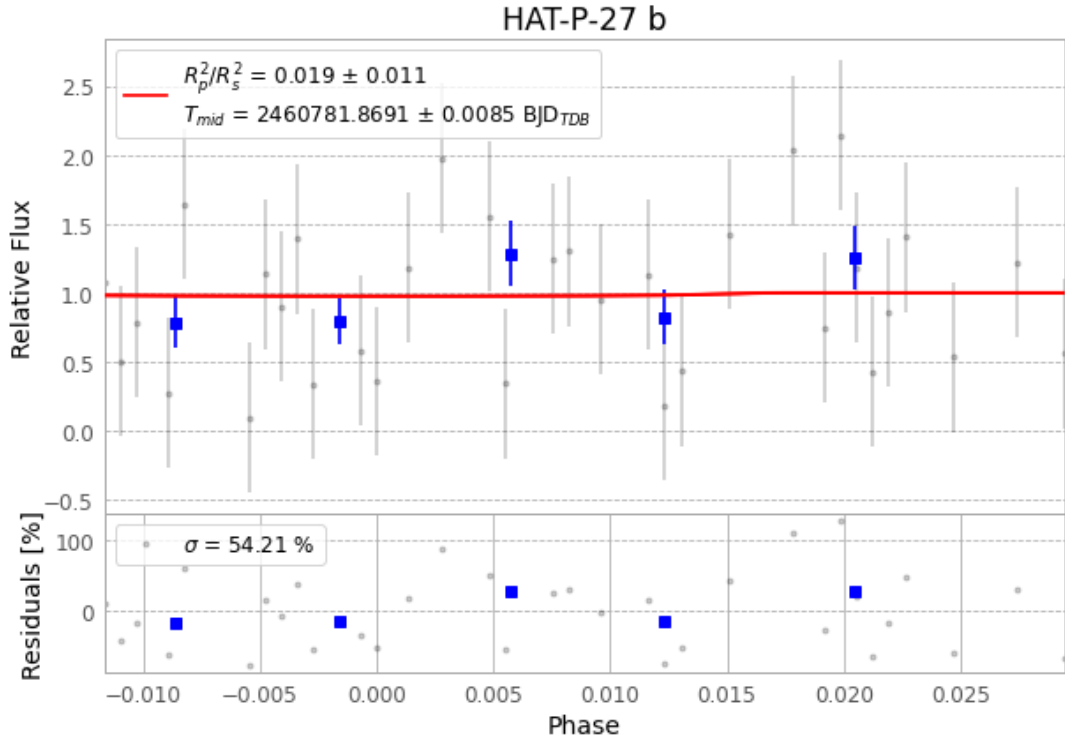


Figure 1 — FinalLightCurve_HAT-P-27 b_2025-04-16.png (SHA-256 fe84f7a35f7d8b21...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [216, 411], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b6b291a56c42b7fa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

64 FITS frames (42 MB), HATP-27250416074515.FITS → HATP-27250416105415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-250416.log (a79e9510cfb8...), FinalLightCurve_HAT-P-27 b_2025-04-16.png (fe84f7a35f7d...), FinalLightCurve_HAT-P-27 b_2025-04-16.csv (b8b09447105e...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 22:21Z.